



PES UNIVERSITY

Department of Computer Science & Engineering

Course Title: Machine Learning

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Machine Learning Laboratory Exercise – 1

Data Quality Assessment and Exploratory Data Analysis on StayEase Booking

Faculty In-charge : Dr. Uma D.

Student In-charge: Chirag

1. Background

StayEase Hotels operates multiple hotel properties across different locations. Over the past year, the management has observed fluctuations in booking demand, increasing booking cancellations, inconsistent room occupancy, and variations in customer behavior.

The management believes that valuable insights can be obtained from historical booking data but lacks a detailed understanding of the underlying factors affecting business performance.

As a **Data Analyst**, you have been assigned to investigate the available booking data and prepare an analytical report that will assist the management in making informed business decisions.

Your responsibility is **not only to analyze the data but also to justify every analytical decision, support conclusions with evidence, and communicate findings in a professional manner.**

2. Purpose of the Study

The purpose of this laboratory exercise is to evaluate the student's ability to investigate a real-world dataset, identify data quality issues, prepare data for analysis, discover meaningful patterns, and communicate business insights through appropriate analytical techniques.

Students are expected to independently determine suitable preprocessing methods, statistical analyses, visualizations, and interpretations based on the characteristics of the dataset.

3. Laboratory Objectives

After completing this laboratory exercise, students should be able to

- investigate a real-world dataset independently.
- identify and resolve data quality issues.
- select suitable statistical and visualization techniques.
- justify analytical decisions using evidence.
- interpret analytical results.
- communicate findings to technical and non-technical audiences.
- recommend business actions based on data.

4. Course Outcomes

Upon successful completion of this laboratory exercise, students will be able to

CO1 Analyze the quality and characteristics of a real-world dataset.

CO2 Apply suitable preprocessing techniques based on identified data quality issues.

CO3 Perform exploratory and statistical analysis to discover meaningful patterns.

CO4 Interpret visual and statistical results to support decision making.

CO5 Present analytical findings and business recommendations in a professional report.

5. Software Requirements

- Google Colab / Jupyter Notebook
- Python
- NumPy
- Pandas
- Matplotlib
- Seaborn
- Plotly (optional)

6. Dataset

Dataset Name: StayEase Booking Analytics Dataset

The dataset contains historical hotel booking information collected from multiple hotel properties. The data includes customer demographics, reservation details, booking channels, pricing information, room allocation, stay characteristics, and related operational information.

The dataset represents real-world business data and may contain imperfections that require investigation before meaningful analysis can be performed.

7. Mission

You have been appointed as a **Junior Data Analyst** at StayEase Hotels.

Your manager has requested an analytical report answering the following question:

"What is happening in our business, why is it happening, and what actions should we take?"

No analytical procedure has been prescribed.

You are expected to determine

- what should be investigated,
- which analytical techniques are appropriate,
- which visualizations best communicate the findings,
- and how the results should be interpreted.

Every conclusion must be supported by evidence.

8. Investigation Requirements

Your investigation should address the following objectives.

Investigation 1

Understand the Dataset

Prepare an overview of the available data.

Your report should include

- description of the dataset
- classification of available attributes
- observations regarding the dataset structure
- important features for business analysis
- possible challenges before analysis begins

Evidence Required

- summary tables
- dataset overview
- supporting observations

Investigation 2

Evaluate Data Readiness

Determine whether the dataset is suitable for business analysis.

Identify every issue that may reduce analytical reliability.

For every issue discovered

- explain why it is important
- describe its impact
- recommend an appropriate treatment
- justify your recommendation

Evidence Required

- summary tables
- analytical observations
- supporting screenshots or outputs

Investigation 3

Prepare the Dataset

Transform the dataset into an analysis-ready form.

Every preprocessing decision must be documented.

For every modification

- identify the problem
- explain the chosen solution
- explain why alternative approaches were not selected
- discuss the impact of the decision

Evidence Required

A preprocessing log similar to the following:

Issue Evidence Action Taken Justification Impact

Investigation 4

Explore Customer and Booking Behaviour

Study customer behaviour using suitable statistical techniques and visualizations.

Do **not** limit yourself to basic descriptive analysis.

Your objective is to discover meaningful patterns.

Your report should include

- important customer behaviours
- unusual observations
- seasonal trends
- pricing patterns
- booking behaviour
- occupancy behaviour

Every observation must be supported by statistical evidence or visualizations.

Investigation 5

Investigate Business Relationships

Determine relationships among important business variables.

Examples include relationships involving

- customer behaviour
- pricing
- booking channels
- booking cancellations
- seasonal demand
- room allocation

You are free to select additional relationships that you consider useful.

Every conclusion must be justified using analytical evidence.

Investigation 6

Discover Business Insights

Identify the most valuable findings obtained during the investigation.

Do not merely describe graphs.

Instead

- explain why the pattern occurs
- explain its business importance
- discuss possible operational impact

Support every insight using evidence.

Investigation 7

Recommend Business Actions

Based on your complete investigation, prepare recommendations for the management.

Your recommendations should be practical and supported by analytical evidence.

Possible areas include

- reducing cancellations
- improving occupancy
- increasing revenue
- customer retention
- pricing
- seasonal planning
- marketing

9. Final Deliverables

Submit the following.

Deliverable 1

Python Notebook

The notebook should demonstrate the complete analytical workflow with appropriate documentation.

Deliverable 2

Analyst's Investigation Report (5–8 pages)

The report should contain

- Executive Summary
- Business Problem
- Investigation Methodology
- Evidence Collected
- Major Findings
- Business Insights
- Recommendations
- Limitations
- Conclusion

Deliverable 3

Decision Log

Maintain a record of every important analytical decision.

| Decision | Reason | Evidence | Alternative Considered |

Deliverable 4

Visualization Portfolio

Prepare a collection of meaningful visualizations.

For every visualization include

- objective
- interpretation
- business significance
- limitations